Homework — 2.1.4 Thinking Logically — ANSWER SHEET

Separate answer sheet / mark scheme — keep for marking.

Mark scheme

Part A (14 marks)

1. (a) [1] `IF height >= 1.4` (accept `>= 140` if cm stated). Must use `>=` for "at least".

(b) [2] 1 mark TRUE: the rider is allowed on / admitted. 1 mark FALSE: the rider is refused / turned away.

2. [2] 1 mark condition: `IF amount <= balance AND amount <= dailyLimit` (accept equivalent variable names). 1 mark FALSE outcome: the withdrawal is **refused/declined** (insufficient funds or over the daily limit) and no cash is dispensed. Must use a logical operator (AND).

3. (a) [2] 1 mark: `IF level < 20` (THEN pump ON). 1 mark: `IF level > 80` (THEN pump OFF). *(Accept* `<=`/`>=` *only if consistent with the wording; strict* `<`/`>` *match "below"/"above".)*

(b) [1] Three outcomes: turn ON, turn OFF, or leave unchanged.

4. [2] 1 mark: under `spent > 100`, a customer who has spent exactly £100 does **not** qualify (100 is not greater than 100). 1 mark: rewrite as `IF spent >= 100`.

5. [4] Up to 4: First decision `IF score >= 70` → TRUE: Distinction (1 for condition, 1 for outcome). Otherwise `IF score >= 40` → TRUE: Pass; ELSE: Fail (1 for the second condition, 1 for both its branch outcomes). Recognise the conditions are **ordered/nested** so 70+ is tested first (credit within the marks above). Conditions must be exact Booleans with both branches stated and applied to this scenario; vague "if it's high" wording earns no condition mark.

Part B (6 marks)

6. [2] 1 mark: the tax step uses the **gross pay** figure produced by the previous step. 1 mark: it therefore depends on that output (a data dependency), so it cannot run until gross pay has been calculated — the order is fixed/sequential.

7. [2] 1 mark: a **primary key** is an attribute (or set of attributes) that **uniquely identifies** each record in a table. 1 mark: a **foreign key** is an attribute in one table that refers to the primary key of another table, linking the two (enforcing referential integrity).

8. [2] `00101101`. 1 mark for correct binary value (101101), 1 mark for correct 8-bit form with leading zeros. *(45 = 32 + 8 + 4 + 1.)*

Total: 14 + 6 = 20 marks